

Tweedie Denoising Step for Diffusion Sampling (PF-ODE and Full SDE)

1 The Problem: Grainy Samples

The forward SDE for the passive model is the OU process

$$dx = -kx dt + \sqrt{2T_p} dW, \quad t \in [0, T], \quad (1)$$

with marginal distribution $p_t(x_t | x_0) = \mathcal{N}(\mu_t, \sigma_t^2)$ where

$$\mu_t = e^{-kt}x_0, \quad \sigma_t^2 = \frac{T_p}{k}(1 - e^{-2kt}). \quad (2)$$

The score is trained with a minimum time $t_{\min} = 10^{-3}$ (i.e. t is sampled uniformly on $[t_{\min}, T]$), so the model has *never seen* $t = 0$ during training.

Both samplers integrate backwards from $t = T$ toward $t = 0$ and call the score network at each step. The last step reaches $t = 0$, which is out of the training distribution, so the model returns a garbage score that corrupts the final sample.

PF-ODE (Heun integrator). The time grid ends at $t_{\text{next}} = 0$. The Heun corrector evaluates the score at both t_{curr} and $t_{\text{next}} = 0$, so the final update is polluted.

Full SDE (Euler–Maruyama). The discrete time grid is $t_i = i \cdot \Delta t$ for $i = N - 1, \dots, 0$. At the last step $i = 0$, $t = 0$: the model is called out of distribution, the resulting drift corrupts $x_{\Delta t}$, and no noise is added to partially mask it.

2 The Fix: Stop at $t_{\min}$ and Apply Tweedie

Step 1 (PF-ODE). Change the time grid to end at $t_{\min} = 10^{-3}$ instead of $t = 0$, so the last Heun corrector evaluation stays within the training range.

Step 1 (SDE). Change the Euler loop to stop at $i = 1$ (i.e. $t = \Delta t$) instead of $i = 0$ ($t = 0$), so the model is never called at $t = 0$.

Step 2 (both). After the loop stops at $x_{t_{\min}}$, apply one *Tweedie denoising step* to obtain a clean estimate of x_0 .

2.1 Tweedie’s Formula

For a Gaussian perturbation kernel $p(x_t \mid x_0) = \mathcal{N}(\mu_t, \sigma_t^2)$, Tweedie’s identity gives the posterior mean:

$$\hat{x}_0 = \mathbb{E}[x_0 \mid x_t] = \frac{x_t + \sigma_t^2 \nabla_x \log p_t(x_t)}{e^{-kt}}. \quad (3)$$

At $t = t_{\min}$:

$$a = e^{-k t_{\min}}, \quad \sigma^2 = \frac{T_p}{k} (1 - e^{-2k t_{\min}}), \quad (4)$$

and one forward pass of the score network at $(x_{t_{\min}}, t_{\min})$ gives $s_\theta \approx \nabla_x \log p_{t_{\min}}(x_{t_{\min}})$. The denoised output is then

$$\hat{x}_0 = \frac{x_{t_{\min}} + \sigma^2 s_\theta(x_{t_{\min}}, t_{\min})}{a}. \quad (5)$$

This requires *one extra NFE* (network function evaluation) but completely removes the residual noise at $t_{\min}$.

3 Why It Works

Because the forward process is Gaussian, $\hat{x}_0$ in (4) is the *minimum mean-squared-error* (MMSE) estimator of x_0 given x_t . Even if the score is only approximately correct, the estimator degrades gracefully: any error in s_θ translates linearly into an error in $\hat{x}_0$.

At $t_{\min} = 10^{-3}$ with typical parameters $k = 4$, $T_p = 6.4$:

$$\sigma = \sqrt{\frac{6.4}{4} (1 - e^{-0.008})} \approx 0.11.$$

Without the Tweedie step, $x_{t_{\min}}$ still carries noise of standard deviation ≈ 0.11 (roughly 8% of the image range), which is perceptible as graininess. The Tweedie step removes this in a single pass.

4 Connection to CLD-SGM

This is exactly the `denoising = true` flag in the CLD-SGM codebase (Dockhorn et al., ICLR 2022). Their `sampling.py` applies the same step after every ODE and EM sampler:

```
if config.denoising:
    u = denoising_fn(model, u, 1. - eps)    # ODE sampler
    _, u = step_fn(model, u, 1.-eps, eps)    # EM sampler
```

where the second return value of `step_fn` is the conditional mean $\hat{x}_0$ (the Tweedie estimate), not the noisy sample.

5 Active Model

For the active model with state (x_t, η_t) , the analogous Tweedie step is applied to the x channels only (since η is discarded at the end):

$$\hat{x}_0 = \frac{x_{t_{\min}} + M_{11}(t_{\min}) F_x(x_{t_{\min}}, \eta_{t_{\min}}, t_{\min})}{e^{-k t_{\min}}}, \quad (6)$$

where $M_{11}(t)$ is the (x, x) element of the joint covariance matrix and F_x is the network’s x -score output. This is exact for the marginal x -score in the limit $\tau \rightarrow 0$, and approximate for finite τ .

References

- [1] T. Dockhorn, A. Vahdat, and K. Kreis, *Score-Based Generative Modeling with Critically-Damped Langevin Diffusion*, ICLR 2022.
- [2] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, *Score-Based Generative Modeling through Stochastic Differential Equations*, ICLR 2021.